Packet Tracer Lab: Single-Area OSPF Office Network

Oct 5, 2026 · @Cory

Overview

In this lab you build a three-floor office and its data center, run single-area OSPF (Area 0) across all of it, and then prove the design is redundant. You power off each data center core router in turn while a workstation keeps pinging the server.

Scenario. Each floor has one router, one switch and one PC. Every floor router has two uplinks, one to each of the two data center core routers (CORE-1 and CORE-2). The data center has three more routers behind the cores:

- **DC-SRV** is the gateway for the server LAN, where the ping-test server lives.
- **DC-EDGE** is the internet gateway. It holds a default route to a simulated ISP and shares that route with the rest of the network through OSPF.
- **DC-MGMT** is the management router. Its loopback stands in for a management network and gives you a second ping target.

By the end of the lab you will have:

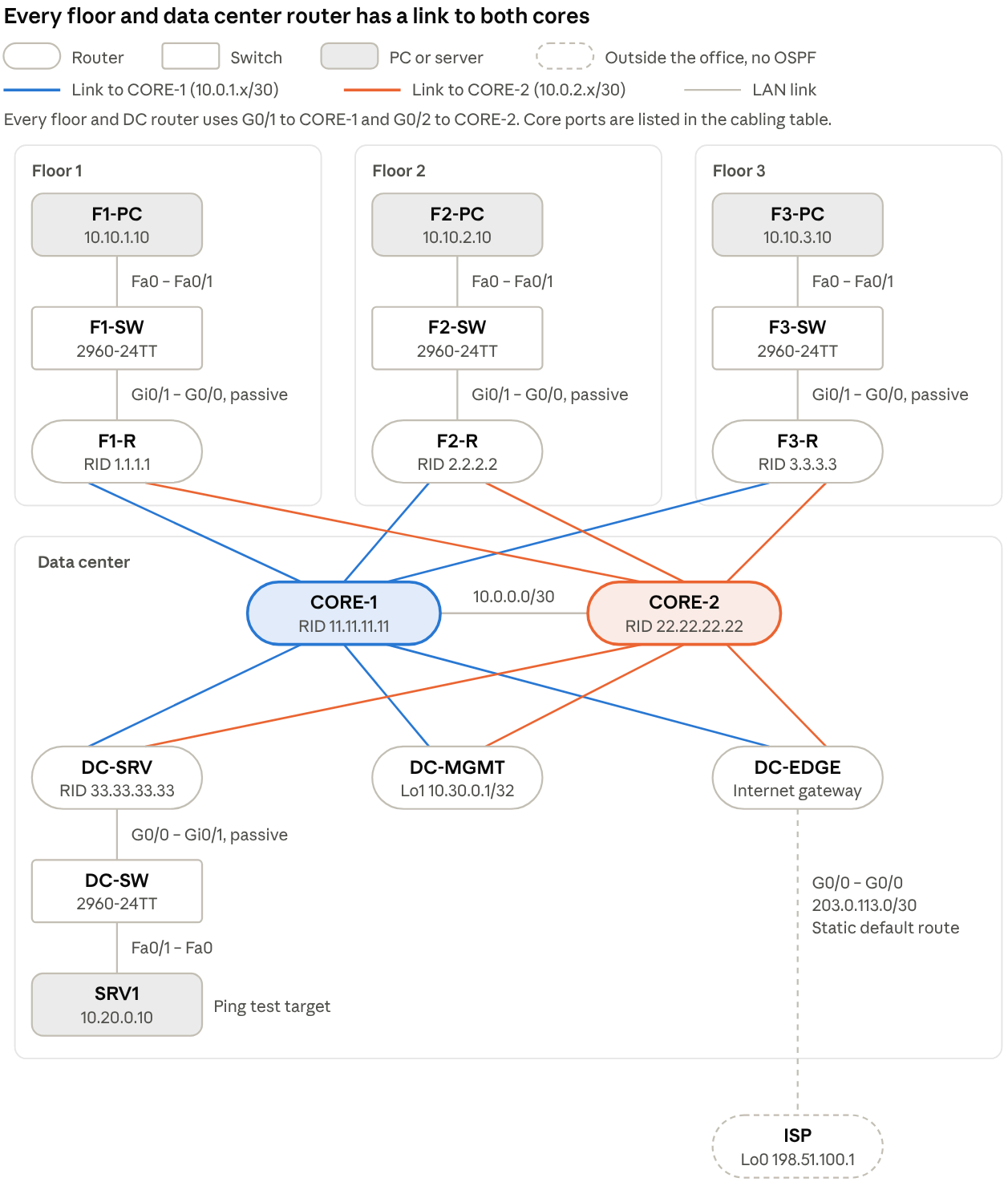
1. Built and cabled a 17-device topology in Packet Tracer, including two modular core routers.
2. Configured OSPF with manual router IDs, a consistent reference bandwidth and precise `network` statements.
3. Made each floor's LAN interface passive, so the floor subnet is advertised but no OSPF Hellos go toward the users.
4. Shared a default route from the internet gateway with `default-information originate`.
5. Verified neighbors, routes and equal-cost paths with `show` commands.
6. Shown that traffic survives the loss of either core router. You will also learn how long a router needs to boot and rebuild its adjacencies before it is safe to take down its partner.

What you need: Cisco Packet Tracer 8.x and about 2 hours: roughly 30 minutes to build, 45 minutes to configure, 15 minutes to verify and 30 minutes for the failover test.

Note: this guide is the lab itself. Packet Tracer stores labs in its own .pkt format, so you build the topology from the tables below and paste in the configurations.

Topology

Each floor router and each data center router has one link to CORE-1 and one to CORE-2, so either core can fail without cutting anything off.



Lab topology · 3 floors, 5 data center routers, simulated ISP

Every router runs OSPF in Area 0 except ISP. CORE-1 and CORE-2 are also linked directly (10.0.0.0/30).

Device	Packet Tracer model	Role	OSPF router ID
F1-R	2911	Floor 1 router	1.1.1.1
F2-R	2911	Floor 2 router	2.2.2.2
F3-R	2911	Floor 3 router	3.3.3.3
CORE-1	Router-PT-Empty + 7 × PT-ROUTER-NM-1CGE	Data center core	11.11.11.11
CORE-2	Router-PT-Empty + 7 × PT-ROUTER-NM-1CGE	Data center core	22.22.22.22
DC-SRV	2911	Server LAN gateway	33.33.33.33
DC-EDGE	2911	Internet gateway	44.44.44.44
DC-MGMT	2911	Management router	55.55.55.55
ISP	2911	Simulated internet provider	Not in OSPF
F1-SW, F2-SW, F3-SW, DC-SW	2960-24TT	Floor and server LAN switches	
F1-PC, F2-PC, F3-PC	PC-PT	Floor workstations	
SRV1	Server-PT	Ping and web test target	

Cabling and addressing

The addressing follows one rule: every transit link to CORE-1 is in 10.0.1.0/24, and every transit link to CORE-2 is in 10.0.2.0/24. The core router takes the first usable address of each /30 and the other router takes the second.

Cabling

Use **Copper Cross-Over** between two routers. Use **Copper Straight-Through** for router-to-switch and switch-to-end-device links.

Device A	Port	Device B	Port	Cable
F1-PC	Fa0	F1-SW	Fa0/1	Straight-through
F1-SW	Gi0/1	F1-R	G0/0	Straight-through
F1-R	G0/1	CORE-1	G0/0	Cross-over
F1-R	G0/2	CORE-2	G0/0	Cross-over
F2-PC	Fa0	F2-SW	Fa0/1	Straight-through
F2-SW	Gi0/1	F2-R	G0/0	Straight-through
F2-R	G0/1	CORE-1	G1/0	Cross-over
F2-R	G0/2	CORE-2	G1/0	Cross-over
F3-PC	Fa0	F3-SW	Fa0/1	Straight-through
F3-SW	Gi0/1	F3-R	G0/0	Straight-through
F3-R	G0/1	CORE-1	G2/0	Cross-over
F3-R	G0/2	CORE-2	G2/0	Cross-over
DC-SRV	G0/1	CORE-1	G3/0	Cross-over
DC-SRV	G0/2	CORE-2	G3/0	Cross-over
DC-EDGE	G0/1	CORE-1	G4/0	Cross-over
DC-EDGE	G0/2	CORE-2	G4/0	Cross-over
DC-MGMT	G0/1	CORE-1	G5/0	Cross-over
DC-MGMT	G0/2	CORE-2	G5/0	Cross-over
CORE-1	G6/0	CORE-2	G6/0	Cross-over
DC-SRV	G0/0	DC-SW	Gi0/1	Straight-through
DC-SW	Fa0/1	SRV1	Fa0	Straight-through
DC-EDGE	G0/0	ISP	G0/0	Cross-over

Addressing

Device	Interface	Address	Connects to	In OSPF
F1-R	G0/0	10.10.1.1/24	Floor 1 LAN	Yes, passive
F1-R	G0/1	10.0.1.2/30	CORE-1	Yes
F1-R	G0/2	10.0.2.2/30	CORE-2	Yes
F2-R	G0/0	10.10.2.1/24	Floor 2 LAN	Yes, passive
F2-R	G0/1	10.0.1.6/30	CORE-1	Yes
F2-R	G0/2	10.0.2.6/30	CORE-2	Yes
F3-R	G0/0	10.10.3.1/24	Floor 3 LAN	Yes, passive
F3-R	G0/1	10.0.1.10/30	CORE-1	Yes
F3-R	G0/2	10.0.2.10/30	CORE-2	Yes
CORE-1	G0/0	10.0.1.1/30	F1-R	Yes
CORE-1	G1/0	10.0.1.5/30	F2-R	Yes
CORE-1	G2/0	10.0.1.9/30	F3-R	Yes
CORE-1	G3/0	10.0.1.13/30	DC-SRV	Yes
CORE-1	G4/0	10.0.1.17/30	DC-EDGE	Yes
CORE-1	G5/0	10.0.1.21/30	DC-MGMT	Yes
CORE-1	G6/0	10.0.0.1/30	CORE-2	Yes
CORE-2	G0/0	10.0.2.1/30	F1-R	Yes
CORE-2	G1/0	10.0.2.5/30	F2-R	Yes
CORE-2	G2/0	10.0.2.9/30	F3-R	Yes
CORE-2	G3/0	10.0.2.13/30	DC-SRV	Yes
CORE-2	G4/0	10.0.2.17/30	DC-EDGE	Yes
CORE-2	G5/0	10.0.2.21/30	DC-MGMT	Yes
CORE-2	G6/0	10.0.0.2/30	CORE-1	Yes

Device	Interface	Address	Connects to	In OSPF
DC-SRV	G0/0	10.20.0.1/24	Server LAN	Yes, passive
DC-SRV	G0/1	10.0.1.14/30	CORE-1	Yes
DC-SRV	G0/2	10.0.2.14/30	CORE-2	Yes
DC-EDGE	G0/0	203.0.113.2/30	ISP	No
DC-EDGE	G0/1	10.0.1.18/30	CORE-1	Yes
DC-EDGE	G0/2	10.0.2.18/30	CORE-2	Yes
DC-MGMT	Lo1	10.30.0.1/32	Simulated management network	Yes
DC-MGMT	G0/1	10.0.1.22/30	CORE-1	Yes
DC-MGMT	G0/2	10.0.2.22/30	CORE-2	Yes
ISP	G0/0	203.0.113.1/30	DC-EDGE	No
ISP	Lo0	198.51.100.1/32	Simulated internet host	No

End devices

Device	IP address	Mask	Default gateway
F1-PC	10.10.1.10	255.255.255.0	10.10.1.1
F2-PC	10.10.2.10	255.255.255.0	10.10.2.1
F3-PC	10.10.3.10	255.255.255.0	10.10.3.1
SRV1	10.20.0.10	255.255.255.0	10.20.0.1

Part 1: Build the topology

You build the topology in four steps: place the devices, give the core routers enough Gigabit ports, name everything, then cable it from the cabling table.

1. **Place the devices.** Drag these onto the Logical workspace:
    - Seven **2911** routers: F1-R, F2-R, F3-R, DC-SRV, DC-EDGE, DC-MGMT and ISP.

- Two **Router-PT-Empty** routers: CORE-1 and CORE-2. These are the last items in the Routers list.
- Four **2960-24TT** switches: F1-SW, F2-SW, F3-SW and DC-SW.
- Three **PC-PT** workstations: F1-PC, F2-PC and F3-PC.
- One **Server-PT**: SRV1.
- Lay it out like the topology diagram: floors stacked on the left, the two cores in the middle, and the data center routers on the right.

2. **Add ports to the core routers.** A 2911 has only three Gigabit ports, but each core needs seven: three floors, three data center routers and the other core.
   a. Click CORE-1 and open the **Physical** tab.
   b. Click the power switch on the chassis to turn the router **off**. Modules can only be inserted while it is off.
   c. Drag a **PT-ROUTER-NM-1CGE** module into each of slots 0 through 6. That gives you GigabitEthernet0/0, 1/0, 2/0, 3/0, 4/0, 5/0 and 6/0.
   d. Turn the power back **on**.
   e. Repeat for CORE-2.
3. **Name every device.** On each device's **Config** tab, set the Display Name to the name in the cabling table. Router hostnames are set by the configurations in Part 2.
4. **Cable the network.** Follow the cabling table exactly, because the configurations assume those ports. Choose the cable type yourself rather than using the automatic connection, so the ports match the table.
5. **Check the link lights.** Router-to-router links stay red until the interfaces are enabled in Part 2. Switch ports go amber, then green after about 30 seconds while spanning tree runs.

Tip: if your version of Packet Tracer has no Router-PT-Empty, use **Router-PT**. Turn it off, remove its default FastEthernet and Serial modules, and add the NM-1CGE modules to slots 0 through 6 the same way.

Part 2: Configure the network

Every router gets the same OSPF settings: process 1, a manual router ID, a reference bandwidth of 10,000 Mbps, and precise `network` statements in Area 0. Each configuration below can be pasted whole into the router's **CLI** tab.

Every configuration ends with `write memory`. **Do not skip it.** Powering a router off in Packet Tracer erases its running configuration. Without a saved startup configuration, the core router you restart in Part 4 comes back blank.

Step 1: End devices

On each PC and on SRV1, open **Desktop > IP Configuration**, choose **Static**, and enter the address, mask and gateway from the end devices table. SRV1's HTTP service is on by default, and you will use it for an extra test in Part 3. The switches need no configuration.

Step 2: Floor routers

G0/0 faces the floor switch. `passive-interface GigabitEthernet0/0` keeps the floor subnet in OSPF but stops Hellos on that port. No neighbor can form toward the users, and the PCs never see OSPF traffic.

F1-R

```
enable
configure terminal
hostname F1-R
no ip domain-lookup
interface GigabitEthernet0/0
 description Floor 1 LAN (F1-SW)
 ip address 10.10.1.1 255.255.255.0
 no shutdown
interface GigabitEthernet0/1
 description Uplink to CORE-1
 ip address 10.0.1.2 255.255.255.252
 no shutdown
interface GigabitEthernet0/2
 description Uplink to CORE-2
 ip address 10.0.2.2 255.255.255.252
 no shutdown
router ospf 1
 router-id 1.1.1.1
 auto-cost reference-bandwidth 10000
 passive-interface GigabitEthernet0/0
 network 10.10.1.0 0.0.0.255 area 0
```

```
 network 10.0.1.0 0.0.0.3 area 0
 network 10.0.2.0 0.0.0.3 area 0
end
write memory
```

F2-R

```
enable
configure terminal
hostname F2-R
no ip domain-lookup
interface GigabitEthernet0/0
 description Floor 2 LAN (F2-SW)
 ip address 10.10.2.1 255.255.255.0
 no shutdown
interface GigabitEthernet0/1
 description Uplink to CORE-1
 ip address 10.0.1.6 255.255.255.252
 no shutdown
interface GigabitEthernet0/2
 description Uplink to CORE-2
 ip address 10.0.2.6 255.255.255.252
 no shutdown
router ospf 1
 router-id 2.2.2.2
 auto-cost reference-bandwidth 10000
 passive-interface GigabitEthernet0/0
 network 10.10.2.0 0.0.0.255 area 0
 network 10.0.1.4 0.0.0.3 area 0
 network 10.0.2.4 0.0.0.3 area 0
end
write memory
```

F3-R

```
enable
configure terminal
hostname F3-R
no ip domain-lookup
interface GigabitEthernet0/0
 description Floor 3 LAN (F3-SW)
 ip address 10.10.3.1 255.255.255.0
 no shutdown
```

```
interface GigabitEthernet0/1
 description Uplink to CORE-1
 ip address 10.0.1.10 255.255.255.252
 no shutdown
interface GigabitEthernet0/2
 description Uplink to CORE-2
 ip address 10.0.2.10 255.255.255.252
 no shutdown
router ospf 1
 router-id 3.3.3.3
 auto-cost reference-bandwidth 10000
 passive-interface GigabitEthernet0/0
 network 10.10.3.0 0.0.0.255 area 0
 network 10.0.1.8 0.0.0.3 area 0
 network 10.0.2.8 0.0.0.3 area 0
end
write memory
```

Step 3: Core routers

All six of CORE-1's links to other routers fall inside 10.0.1.0/24, so one `network 10.0.1.0 0.0.0.255` statement enables OSPF on all of them. A second statement covers the link between the cores. CORE-2 does the same with 10.0.2.0/24.

CORE-1

```
enable
configure terminal
hostname CORE-1
no ip domain-lookup
interface GigabitEthernet0/0
 description To F1-R
 ip address 10.0.1.1 255.255.255.252
 no shutdown
interface GigabitEthernet1/0
 description To F2-R
 ip address 10.0.1.5 255.255.255.252
 no shutdown
interface GigabitEthernet2/0
 description To F3-R
 ip address 10.0.1.9 255.255.255.252
 no shutdown
interface GigabitEthernet3/0
```

```
 description To DC-SRV
 ip address 10.0.1.13 255.255.255.252
 no shutdown
interface GigabitEthernet4/0
 description To DC-EDGE
 ip address 10.0.1.17 255.255.255.252
 no shutdown
interface GigabitEthernet5/0
 description To DC-MGMT
 ip address 10.0.1.21 255.255.255.252
 no shutdown
interface GigabitEthernet6/0
 description To CORE-2
 ip address 10.0.0.1 255.255.255.252
 no shutdown
router ospf 1
 router-id 11.11.11.11
 auto-cost reference-bandwidth 10000
 network 10.0.1.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
end
write memory
```

CORE-2

```
enable
configure terminal
hostname CORE-2
no ip domain-lookup
interface GigabitEthernet0/0
 description To F1-R
 ip address 10.0.2.1 255.255.255.252
 no shutdown
interface GigabitEthernet1/0
 description To F2-R
 ip address 10.0.2.5 255.255.255.252
 no shutdown
interface GigabitEthernet2/0
 description To F3-R
 ip address 10.0.2.9 255.255.255.252
 no shutdown
interface GigabitEthernet3/0
 description To DC-SRV
```

```
 ip address 10.0.2.13 255.255.255.252
 no shutdown
interface GigabitEthernet4/0
 description To DC-EDGE
 ip address 10.0.2.17 255.255.255.252
 no shutdown
interface GigabitEthernet5/0
 description To DC-MGMT
 ip address 10.0.2.21 255.255.255.252
 no shutdown
interface GigabitEthernet6/0
 description To CORE-1
 ip address 10.0.0.2 255.255.255.252
 no shutdown
router ospf 1
 router-id 22.22.22.22
 auto-cost reference-bandwidth 10000
 network 10.0.2.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
end
write memory
```

Step 4: Data center routers

DC-SRV's server LAN interface is passive for the same reason the floor interfaces are. DC-EDGE has a static default route toward the ISP, and `default-information originate` shares that route with every other router through OSPF. Its ISP-facing port is left out of OSPF entirely.

DC-SRV

```
enable
configure terminal
hostname DC-SRV
no ip domain-lookup
interface GigabitEthernet0/0
 description Server LAN (DC-SW)
 ip address 10.20.0.1 255.255.255.0
 no shutdown
interface GigabitEthernet0/1
 description Uplink to CORE-1
 ip address 10.0.1.14 255.255.255.252
 no shutdown
```

```
interface GigabitEthernet0/2
 description Uplink to CORE-2
 ip address 10.0.2.14 255.255.255.252
 no shutdown
router ospf 1
 router-id 33.33.33.33
 auto-cost reference-bandwidth 10000
 passive-interface GigabitEthernet0/0
 network 10.20.0.0 0.0.0.255 area 0
 network 10.0.1.12 0.0.0.3 area 0
 network 10.0.2.12 0.0.0.3 area 0
end
write memory
```

DC-EDGE

```
enable
configure terminal
hostname DC-EDGE
no ip domain-lookup
interface GigabitEthernet0/0
 description Internet (ISP)
 ip address 203.0.113.2 255.255.255.252
 no shutdown
interface GigabitEthernet0/1
 description Uplink to CORE-1
 ip address 10.0.1.18 255.255.255.252
 no shutdown
interface GigabitEthernet0/2
 description Uplink to CORE-2
 ip address 10.0.2.18 255.255.255.252
 no shutdown
ip route 0.0.0.0 0.0.0.0 203.0.113.1
router ospf 1
 router-id 44.44.44.44
 auto-cost reference-bandwidth 10000
 network 10.0.1.16 0.0.0.3 area 0
 network 10.0.2.16 0.0.0.3 area 0
 default-information originate
end
write memory
```

DC-MGMT

```
enable
configure terminal
hostname DC-MGMT
no ip domain-lookup
interface Loopback1
 description Simulated management network
 ip address 10.30.0.1 255.255.255.255
interface GigabitEthernet0/1
 description Uplink to CORE-1
 ip address 10.0.1.22 255.255.255.252
 no shutdown
interface GigabitEthernet0/2
 description Uplink to CORE-2
 ip address 10.0.2.22 255.255.255.252
 no shutdown
router ospf 1
 router-id 55.55.55.55
 auto-cost reference-bandwidth 10000
 network 10.30.0.1 0.0.0.0 area 0
 network 10.0.1.20 0.0.0.3 area 0
 network 10.0.2.20 0.0.0.3 area 0
end
write memory
```

Step 5: Simulated ISP

The ISP router does not run OSPF. Its loopback acts as an internet host, and a static route sends replies for the office's 10.0.0.0/8 addresses back to DC-EDGE.

ISP

```
enable
configure terminal
hostname ISP
no ip domain-lookup
interface Loopback0
 description Simulated internet host
 ip address 198.51.100.1 255.255.255.255
interface GigabitEthernet0/0
 description To office DC-EDGE
 ip address 203.0.113.1 255.255.255.252
 no shutdown
ip route 10.0.0.0 255.0.0.0 203.0.113.2
end
write memory
```

Part 3: Verify OSPF

The network is ready for the failover test when four things are true:

- Each core has 7 FULL neighbors.
- Each floor router has 2 FULL neighbors.
- The floor routers reach the server through both cores at equal cost.
- Every PC can ping SRV1.

Wait about a minute after the last configuration before you check. OSPF waits 40 seconds to elect a DR on each Ethernet link before the neighbors reach FULL.

#	Run this	On	What you should see
1	`show ip ospf neighbor`	CORE-1 and CORE-2	7 neighbors (1.1.1.1, 2.2.2.2, 3.3.3.3, 33.33.33.33, 44.44.44.44, 55.55.55.55 and the other core), all FULL/DR or FULL/BDR
2	`show ip ospf neighbor`	F1-R, F2-R, F3-R	2 neighbors, 11.11.11.11 and 22.22.22.22, both FULL

#	Run this	On	What you should see
3	`show ip protocols`	F1-R	Router ID 1.1.1.1, three networks, and GigabitEthernet0/0 listed under Passive Interface(s)
4	`show ip ospf interface GigabitEthernet0/0`	F1-R	"No Hellos (Passive interface)" and a neighbor count of 0
5	`show ip route ospf`	F1-R	Two equal-cost paths, via 10.0.1.1 and via 10.0.2.1, to 10.20.0.0/24 [110/30], 10.10.2.0/24 [110/30], 10.10.3.0/24 [110/30] and 10.30.0.1/32 [110/21]
6	`show ip route`	F1-R	O*E2 0.0.0.0/0 [110/1], the default route learned from DC-EDGE, through both cores
7	`ping 10.20.0.10`	F1-PC, F2-PC, F3-PC (Command Prompt)	Replies from the server. The first reply may time out while ARP resolves
8	`ping 10.30.0.1` and `ping 198.51.100.1`	F1-PC	Replies from DC-MGMT and from the simulated internet host
9	`tracert 10.20.0.10`	F1-PC	4 hops: 10.10.1.1, then 10.0.1.1 or 10.0.2.1, then 10.0.1.14 or 10.0.2.14, then 10.20.0.10
10	Web Browser to `http://10.20.0.10`	F2-PC (Desktop)	The SRV1 default web page loads

Reading the costs. Every Gigabit link costs 10 with the 10,000 Mbps reference bandwidth. F1-PC to the server is F1-R's uplink (10), plus the core's link to DC-SRV (10), plus DC-SRV's LAN interface (10), for 30. The path through CORE-1 and the path through CORE-2 both cost 30, so OSPF installs both and load-shares. That equal-cost pair is the redundancy Part 4 tests. The management loopback costs 21 because a loopback interface always costs 1.

If a check fails, use the troubleshooting table at the end before you go on. Part 4 only makes sense on a network that passes all ten checks.

Part 4: Failover test

You power off CORE-1 while F1-PC pings the server continuously, and traffic should carry on through CORE-2 with at most a dropped ping or two. Then you bring CORE-1 back, wait until it is fully back in OSPF, and repeat the test with CORE-2.

The waiting matters. A router that has just finished booting is not yet carrying traffic. Its interfaces have to come up, and on each Ethernet link OSPF waits 40 seconds before electing a DR and moving the neighbors to FULL. Power off the second core too early and the floors lose both paths at once. So each restart below ends with a **gate**: checks that must pass before you continue, whatever the clock says.

Before you start

1. On CORE-1 and CORE-2, run `show startup-config` and confirm the full configuration is there. If either shows no saved configuration, run `write memory` on it now.
2. On F1-PC, open **Desktop > Command Prompt** and start a continuous ping to the server: `ping -t 10.20.0.10`. If your version does not accept `-t`, use `ping -n 1000 10.20.0.10`.
3. Optionally, on F3-PC, start `ping -t 198.51.100.1` to watch internet traffic through the same test.
4. Stay in **Realtime** mode. Keep F1-PC's window and F1-R's CLI where you can see them both.

Stage A: Take down CORE-1

1. On F1-R, run `show ip route 10.20.0.0` and record both next hops: 10.0.1.1 (CORE-1) and 10.0.2.1 (CORE-2).
2. Open CORE-1's **Physical** tab and click the power switch to turn it off. Every link to CORE-1 turns red.
3. Watch F1-PC. You may see one or two "Request timed out" lines, then the replies continue.
4. On F1-R, run `show ip ospf neighbor`. Only 22.22.22.22 (CORE-2) is left.
5. On F1-R, run `show ip route 10.20.0.0`. One path remains, via 10.0.2.1.

6. On CORE-2, run `show ip ospf neighbor`. It has 6 neighbors, because the link to CORE-1 is gone.

The network recovers in about a second instead of after the 40-second dead timer. Powering off CORE-1 takes the link down on each neighbor's interface, and a router drops an adjacency as soon as its interface goes down.

Stage B: Bring CORE-1 back

1. Turn CORE-1's power switch back **on**. It boots and loads its saved configuration. If its prompt comes back as `Router>`, the configuration was not saved: paste it again, run `write memory`, and do not continue.
2. Wait at least **2 minutes**. You can click **Fast Forward Time** at the bottom of the window to speed up the Packet Tracer clock.
3. **Gate. Do not go on until all three checks pass:**
   - On CORE-1, `show ip interface brief` shows all seven Gigabit interfaces up/up.
   - On CORE-1, `show ip ospf neighbor` lists all **7** neighbors as FULL. Neighbors in INIT, 2WAY, EXSTART or LOADING are still forming, so wait and check again.
   - On F1-R, `show ip route 10.20.0.0` shows **both** next hops again, 10.0.1.1 and 10.0.2.1.

Stage C: Take down CORE-2

1. With the ping still running, turn CORE-2's power switch **off**.
2. Watch F1-PC. As before, expect at most one or two timeouts, then the replies continue.
3. On F1-R, `show ip ospf neighbor` shows only 11.11.11.11 (CORE-1), and `show ip route 10.20.0.0` shows one path, via 10.0.1.1.

Stage D: Restore and finish

1. Turn CORE-2 back **on** and wait for the same gate: on CORE-2, all seven interfaces up/up and 7 FULL neighbors; on F1-R, two next hops again.
2. Stop the ping on F1-PC with **Ctrl+C** and note the summary line of packets sent, received and lost.

Record your results

Event	Pings lost on F1-PC	F1-R next hops to 10.20.0.0/24	FULL neighbors on the core still up
Baseline, both cores up			
CORE-1 powered off			
CORE-1 back, gate passed			
CORE-2 powered off			
CORE-2 back, gate passed			

Review questions and troubleshooting

Review questions

1. G0/0 on F1-R is passive, yet CORE-2 still has a route to 10.10.1.0/24. What does `passive-interface` stop, and what does it leave alone?
2. Suppose you left out `passive-interface` and someone plugged an unauthorized router into F1-SW. What could that router do to the network?
3. Failover in Part 4 took about a second, not the 40-second dead interval. Why? How would the result change if the floor routers and cores were connected through a switch instead of directly?
4. Why must you wait for the gate before powering off the second core? What do the floors lose if you don't?
5. F1-R reaches 10.20.0.0/24 through two next hops at cost 30. What would its routing table show after you add `ip ospf cost 50` to F1-R's G0/2?
6. Why does the default route appear as O*E2, not as O?
7. What happens to F1-PC's pings if both cores are off at the same time? What single change to the topology would remove that risk?

Troubleshooting

Symptom	Likely cause	Fix
A router-to-router link stays red	Straight-through cable, or the interface is still shut down	Replace it with a cross-over cable; run `no shutdown` on both ends
A core shows fewer than 7 neighbors	Cable in the wrong port, or a mistyped IP address	Compare `show ip interface brief` and `show cdp neighbors` with the cabling and addressing tables
A floor router has no OSPF neighbors	A `network` statement doesn't match the uplink addresses	Check the networks listed in `show ip protocols` against the addressing table
A floor subnet is missing from other routers' tables	The LAN interface isn't covered by a `network` statement	Add `network 10.10.x.0 0.0.0.255 area 0`. Passive interfaces still need one
A PC can ping its gateway but not the server	The PC's or SRV1's default gateway is wrong, or the server LAN isn't advertised	Check IP Configuration on both ends; on any router, `show ip route 10.20.0.0`
The server answers but 198.51.100.1 does not	DC-EDGE is missing the static default or `default-information originate`, or ISP has no return route	Check `show ip route` on DC-EDGE and on ISP
A router has no route to another router's networks, or reports a duplicate router ID	Two routers were given the same router ID	Fix the `router-id`, then run `clear ip ospf process` and answer yes
A core comes back as `Router>` after power-on	The configuration was never saved	Paste the configuration again and run `write memory`
All pings fail when the second core goes off	The first core was not FULL again yet	Restore both cores, wait for the gate, and repeat the stage